

Today

- Linear Systems of Equations
- Gaussian Elim.
- Partial Pivoting

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ -x_1 + x_2 &= 1 \end{aligned}$$

Unknowns: $x_1, \dots, x_n$

Linear Eqn.: $\sum_{j=1}^n a_{ij} x_j = b_i$

no nonlinear functions of x_j
 $x^2, e^x, \log(x), \sin(x)$ etc.

System of m
linear equations

$$\sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, \dots, m$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Book: $[A] \{x\} = \{b\}$

Handwriting: $A \vec{x} = \vec{b}$

for now $m = n$

HWO $\Rightarrow A \cdot x$

Lab

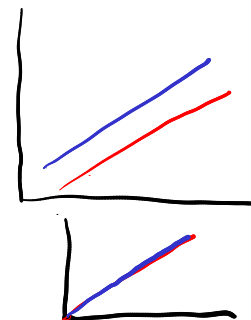
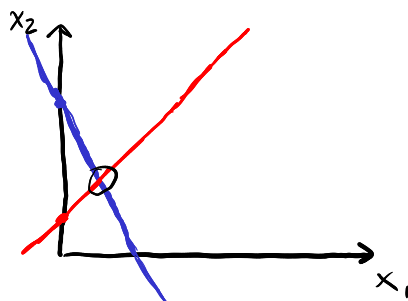
$Ax = b$
 Known $\uparrow$ Unknowns $\uparrow$ Known

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ x_2 &= -2x_1 + 4 \end{aligned}$$

$$-x_1 + x_2 = 1$$

$$x_2 = x_1 + 1$$

$$\boxed{x_1 = 1, x_2 = 2}$$



Example:

$$\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$3x_1 - x_2 = 0 \rightarrow 3x_1 - 3 = 0$$

$$\begin{aligned} 2x_2 &= 6 \\ x_2 &= 3 \end{aligned}$$

$$\boxed{x_1 = 1}$$

$$2x_1 + 3x_2 + 4x_3 = 19 \quad (1)$$

$$4x_1 + 11x_2 + 14x_3 = 55 \quad (2)$$

$$2x_1 + 8x_2 + 17x_3 = 50 \quad (3)$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 4 & 19 \\ 4 & 11 & 14 & 55 \\ 2 & 8 & 17 & 50 \end{array} \right]$$

$$\begin{array}{l} -2(1) \\ -1(1) \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 4 & 19 \\ 0 & 5 & 6 & 17 \\ 0 & 5 & 13 & 31 \end{array} \right]$$

$$-1(2)$$

$$\left[\begin{array}{ccc|c} 2x_1 + 3x_2 + 4x_3 = 19 \\ 0 \quad 5x_2 + 6x_3 = 17 \\ 0 \quad 0 \quad 7x_3 = 14 \end{array} \right]$$

$$x_3 = 2$$

$$2x_1 + 3(1) + 4(2) = 19$$

$$2x_1 = 8$$

$$x_1 = 4$$

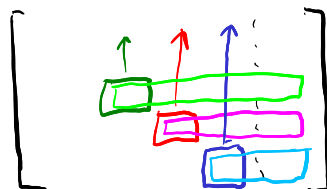
$$5x_2 + 12 = 17$$

$$x_2 = 1$$



Eliminate $\square$ with $\square$
Eliminate $\square$ with $\square$
etc

Back substitution

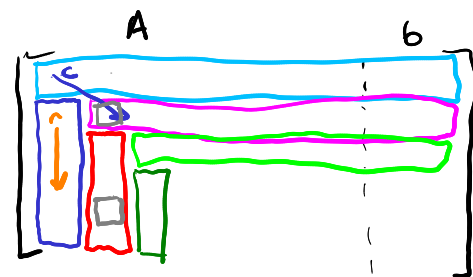


Solve for $\square$ with $\square$
Solve for $\square$ with $\square$
etc.

Example :

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 9 & -2 \\ 2 & 1 & -1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 9 \\ 2 \end{bmatrix}$$

$$x = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$



function Gauss (A, b)

$n \leftarrow \text{length}(b)$

$\text{Aug} \leftarrow [A \mid b]$

for c in $1:n$

for r in $c+1:n$

$f \leftarrow \text{Aug}[r,c] / \text{Aug}[c,c]$

$\text{Aug}[r,c:end] \leftarrow \text{Aug}[r,c:end] - f \cdot \text{Aug}[c,c:end]$

$x \leftarrow \text{zeros}(n,1)$

$x[n] \leftarrow \text{Aug}[n:end] / \text{Aug}[n,n]$

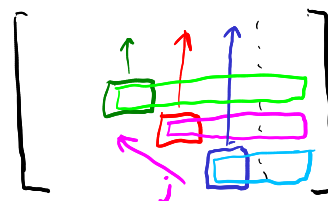
for j in $n-1:-1:1$

$b_j \leftarrow \text{Aug}[j,end]$

$x[j] \leftarrow (b_j - \text{Aug}[j,j+1:n] \cdot x[j+1:n]) / \text{Aug}[j,j]$

return x

Back substitution



dot product
of all x's we've solved for

$$A = \begin{bmatrix} 0 & 2 & 3 \\ 4 & 6 & 7 \\ 2 & -3 & 6 \end{bmatrix} \quad b = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix}$$

$$f \leftarrow \text{Aug}[1,2] / \text{Aug}[1,1]$$

Divide by Zero

Solution: Partial Pivoting

function Gauss^{Pivot}(A, b)

$n \leftarrow \text{length}(b)$

$\text{Aug} \leftarrow [A \mid b]$

for c in $1:n$

$\text{best} \leftarrow \text{argmax}(\text{abs}(\text{Aug}[c:\text{end}, c])) + c - 1$

$\text{Aug}([best, c], :) \leftarrow \text{Aug}[c, best, :]$

for r in $c+1:n$

$f \leftarrow \text{Aug}[r, c] / \text{Aug}[c, c]$

$\text{Aug}[r, c:\text{end}] \leftarrow \text{Aug}[r, c:\text{end}] - f \cdot \text{Aug}[c, c:\text{end}]$

$x \leftarrow \text{zeros}(n, 1)$

$x[n] \leftarrow \text{Aug}[n:\text{end}] / \text{Aug}[n, n]$

for j in $n-1:-1:1$

$b_j \leftarrow \text{Aug}[j, \text{end}]$

$x[j] \leftarrow (b_j - \text{Aug}[j, j+1:n] \cdot x[j+1:n]) / \text{Aug}[j, j]$

dot product
of all x 's we've solved for

return x